

Exercise 1.5

36) Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase "It is not the case that.")

a) No one has lost more than one thousand dollars playing the lottery.

$L(x, y)$: x has lost more than y dollars playing the lottery.

Domain of x : Set of all people

Domain of y : Set of all ~~whole~~ non-negative reals

$\forall x \neg L(x, 1000) \equiv \neg \forall x L(x, 1000) \equiv \exists x \neg L(x, 1000)$ (De Morgan's Law of Quantifiers)

$\neg \forall x \neg L(x, 1000) \equiv \exists x L(x, 1000)$ (By Double-negation Law)

There exists at least one person in the class who has lost more than one thousand dollars while playing the lottery.

b) There is a student in this class who has chatted with exactly one other student.

$C(x, y)$: x has chatted with y . Common Domain: Students of the class

$\exists x \exists y ((x \neq y) \wedge C(x, y) \wedge \forall z ((x \neq z) \wedge C(x, z) \Rightarrow (z = y)))$

$\neg \exists x \exists y ((x \neq y) \wedge C(x, y) \wedge \forall z ((x \neq z) \wedge C(x, z) \Rightarrow (z = y)))$ (De Morgan's Law of quantifiers)

$\equiv \forall x \neg \exists y ((x \neq y) \wedge C(x, y) \wedge \forall z ((x \neq z) \wedge C(x, z) \Rightarrow (z = y)))$ (De Morgan's Law of quantifiers)

$\equiv \forall x \forall y \neg ((x \neq y) \wedge C(x, y) \wedge \forall z ((x \neq z) \wedge C(x, z) \Rightarrow (z = y)))$ (1st De Morgan's Law)

$\equiv \forall x \forall y (\neg(x \neq y) \vee \neg C(x, y) \vee \neg \forall z ((x \neq z) \wedge C(x, z) \Rightarrow (z = y)))$ (De Morgan's Law of quantifiers)

$\equiv \forall x \forall y ((x = y) \vee \neg C(x, y) \vee \exists z (\neg((x \neq z) \wedge C(x, z)) \vee (z = y)))$ (Conditional disjunction equivalence)

$\equiv \forall x \forall y ((x = y) \vee \neg C(x, y) \vee \exists z (\neg(\neg(x \neq z) \wedge C(x, z))) \vee (z = y))$ (2nd De Morgan's Law)